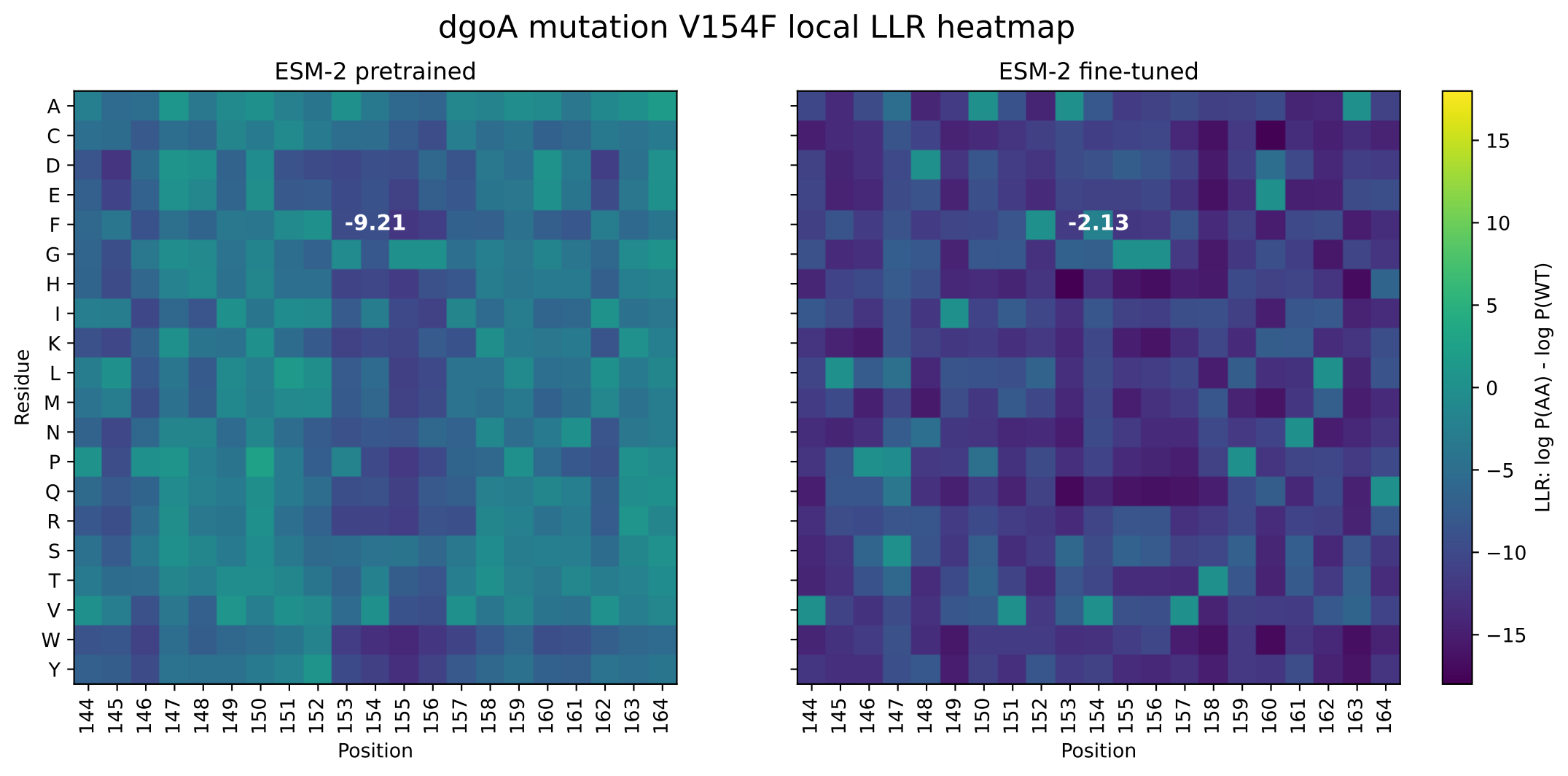
dgoA mutation V154F local LLR heatmap
ESM-2 pretrained
ESM-2 fine-tuned
-9.21
-2.13
Residue
A
C
D
E
F
G
H
I
K
L
M
N
P
Q
R
S
T
V
W
Y
144
145
146
147
148
149
150
151
152
153
154
155
156
157
158
159
160
161
162
163
164
Position
15
10
5
0
−5
−10
−15
LLR: log P(AA) - log P(WT)